BASELINE MODEL REPORT

Predicting Three-Axis Accelerometer Error from Thermal Sensor Data

An informal preliminary analysis

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Background & Question (Recap)

Research question. To what extent can thermal measurements from multiple locations on the sensor package predict or compensate for three-axis accelerometer measurement error over time?

Hypothesis and prediction. Accelerometer error changes systematically with the thermal state of the device. I therefore expect temperature levels, temperature change rates, spatial temperature gradients, and recent temporal context to predict x-, y-, and z-axis error better than a naive mean prediction. Because thermal effects evolve gradually, sequence-aware or autoregressive models should outperform models that use temperature at a single instant.

Methods

Data and feature engineering. The notebook concatenated 4,440,384 observations, converted time to integer seconds, sorted the records, and selected the longest uninterrupted one-second run (746,265 observations). The three targets were x, y, and z acceleration error (m/s^2). Twelve temperature channels were used as predictors. Current temperature values and lags of 1, 2, 5, and 10 samples were created, resulting in 60 temperature-only features and 746,255 complete rows. The autoregressive random forest expanded the matrix to 75 features. The LSTM used 30-step sequences and 33 engineered predictors: 12 temperatures, 12 first differences, three spatial summaries, two sensor-pair differences, and four calendar/cyclical features.

Baseline rationale. K-nearest-neighbors (KNN) provided a simple nonlinear, local benchmark. Random forest supplied a flexible tree-based comparison that can represent nonlinear interactions. The autoregressive random forest tested whether recent acceleration history carried the missing signal. LSTM variants were exploratory sequence baselines intended to learn longer temporal structure.

Training and validation. All main train/test partitions were chronological: the first 80% for training and the last 20% for testing (597,004/149,251 for the classical models; 596,988/149,247 sequences for the LSTM). KNN and random-forest pipelines used five-fold TimeSeriesSplit cross-validation inside GridSearchCV. Standardization and PCA were contained in the pipelines. The LSTM used a further 20% validation split from the training sequences, 20% dropout, and early stopping with patience of three epochs.

Model Assumptions and Checks

Assumption / concern	Evidence in notebook
Chronological ordering and no future leakage	Chronological splits and TimeSeriesSplit were used. However, the LSTM MinMaxScaler was fitted before the split. More importantly, the autoregressive feature list includes current x, y, and z acceleration while predicting those same values; this is target leakage and invalidates that model as a clean benchmark.
Comparable local neighborhoods (KNN)	Standardization and PCA support distance-based comparison. The highly correlated temperature channels (mean inter-sensor correlation = 0.9028) justify dimension reduction, but temporal drift can still make training neighbors unrepresentative of the holdout.
Stable train-to-test relationship	Strongly negative holdout R^2 values reject this assumption in practice: patterns learned earlier did not generalize to the final time block.
Independent residuals / adequate temporal structure	Not formally tested. The targets themselves have lag-1 correlations of 0.999964, 0.999974, and 0.999994, so residual autocorrelation is a major risk. Durbin-Watson or Ljung-Box tests and residual ACF plots are required next.
Homoscedastic, centered errors	Not formally tested. Residual-versus-fitted and residual-versus-time plots, reported separately for each axis, are still needed. These are diagnostics rather than strict distributional requirements for KNN/RF/LSTM.
Feature and target scaling	KNN/RF pipelines used StandardScaler; the LSTM used MinMax scaling for X. Targets were not scaled, which can cause the larger-variance z-axis to dominate a shared MSE loss.

Results

Fitted models

Model	Selected configuration	CV R^2	Test R^2	Test MAE
KNN + PCA	11 neighbors; PCA 99%	-3.1941	-28.1517	1.35×10^{-4}
Random forest + PCA	50 trees; depth 10; PCA 95%	-3.8097	-19.6667	1.05×10^{-4}
AR RF + PCA*	100 trees; depth 25; PCA 95%	-0.5343	-0.5361	2.5865×10^{-5}
LSTM	64 units; dropout 0.20	—	-14.4014	1.6214×10^{-4}
LSTM, separate heads	3 one-unit heads	—	-7.8461	1.7774×10^{-4}
LSTM + attention	4 heads; key dim 16	—	-12.2849	1.8020×10^{-4}

*Table 1. Saved notebook results. Negative R^2 means worse than the test-set mean. *AR RF includes current targets and is not leakage-free.*

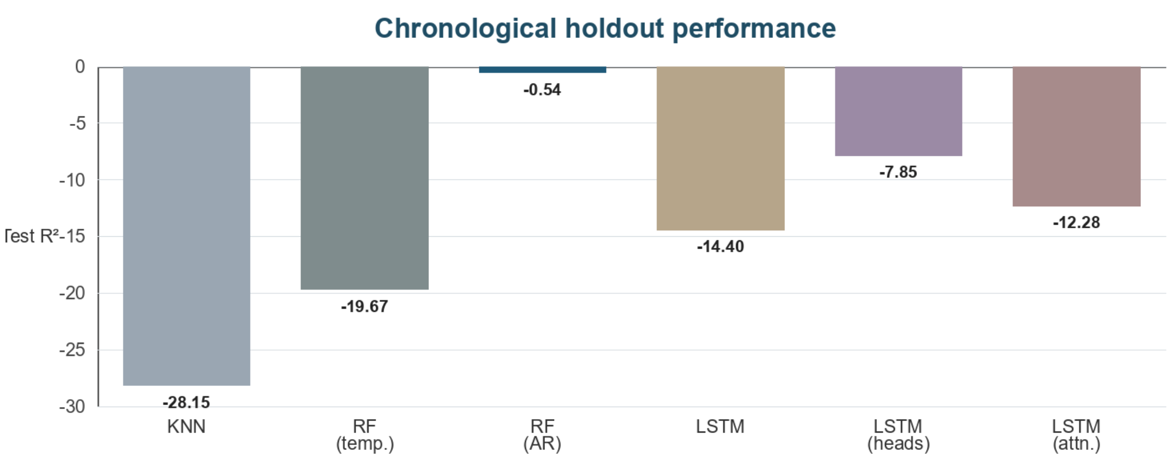

Figure 1. Every tested model had negative R^2 on the chronological holdout.

Interpretation. The temperature-only random forest reduced MAE by about 22% relative to KNN and improved R^2 by 30.1% on the notebook's relative calculation, but both models failed badly out of time. The autoregressive RF has the least-negative saved R^2 (-0.5361) and lowest MAE (2.5865×10^{-5}), but its feature matrix contains current x, y, and z acceleration while predicting those same targets. Its apparent advantage is therefore not a valid estimate of lag-based forecasting performance. The LSTM variants also failed to generalize; the best LSTM R^2 was -7.8461, from the separate-output-head model.

Assumptions and diagnostics

The exploratory statistics support temporal modeling but also show why evaluation must be strict. Lag-1 target autocorrelation exceeded 0.99996 on every axis. Temperature-to-z correlations were strong (for example, r = -0.95093 for T z), while x and y relationships were weaker. The temperature channels were themselves highly redundant (mean cross-correlation 0.9028; range 0.7534–0.9984). These values justify temporal splits and feature reduction, but correlation alone does not establish stable out-of-time prediction.

The notebook did not save residual plots, per-axis metrics, Durbin-Watson/Ljung-Box statistics, or residual normality/variance diagnostics. I therefore cannot claim that residual independence, constant variance, or unbiased errors were satisfied. For these nonlinear predictive models, normality is not required for fitting, but centered errors, stable variance, and absence of remaining temporal structure are still important for interpretation.

Overfitting

Overfitting was controlled partly through a chronological holdout, five-fold time-series cross-validation for KNN/RF, PCA, bounded tree depth, dropout, and early stopping. KNN and temperature-only RF show large negative CV and test R^2 values, so they do not generalize out of time. The LSTM validation loss also deteriorated after its best early epoch. The autoregressive RF result cannot be interpreted as a valid overfitting check until current-time acceleration columns are removed: only lagged target values may be used. In the next run, preprocessing must be fit inside each training fold and sequence windows must not cross split boundaries.

Discussion & Next Steps

Key takeaway. The current results do not support the prediction that the engineered temperature features can yet forecast all three acceleration-error axes better than a naive baseline. The target series are extremely persistent, but the apparently strongest autoregressive random forest includes current-time acceleration columns and therefore cannot establish legitimate predictive skill. The analysis shifts the question from “Can temperature alone predict error?” to “Does temperature add out-of-time predictive value beyond a leakage-free persistence or lag-only autoregressive baseline?”

Revision to the analysis plan. I would retain KNN and temperature-only RF as reference baselines, but make persistence, rolling mean, and lag-only regularized autoregression the primary baselines. I would then measure the incremental value of thermal variables through ablation: lagged acceleration only; temperature only; and both together. Results should be reported per axis because z has much larger variance and much stronger temperature correlations than x or y.

Hyperparameter Tuning & Model Evaluation Plan

- Preserve the corrected integer-second timestamp workflow and document the selected continuous run (746,265 seconds, approximately 8.64 days).
- Use walk-forward or blocked nested cross-validation with a gap/embargo at least as long as the largest lag or sequence window. Fit scalers/PCA only on each training fold.
- Add naive baselines: last observation, rolling mean, seasonal/persistence variants, and per-axis mean prediction.
- Tune autoregression lags/window sizes; random-forest depth, minimum leaf size, maximum features, and tree count; and regularized linear models (Ridge/Elastic Net).
- For LSTM/GRU models, tune sequence length, hidden units, learning rate, batch size, dropout, and weight decay. Scale each target or use per-axis loss weights.
- Evaluate per-axis and macro-average MAE, RMSE, and R^2, plus residual-versus-time/fitted plots, ACF/PACF, Ljung-Box tests, and error by temperature range/time block.
- Use ablation and permutation importance to determine whether temperature contributes beyond acceleration history. Select the final model only on validation data, then evaluate once on an untouched final test block.

Data Processing and Reproducibility

This notebook version consistently converts timestamps to integer seconds and identifies a longest uninterrupted run of 746,265 one-second observations. Two implementation changes are still needed before final modeling. First, create uninterrupted_df with .copy() before dropping columns or adding error_array to avoid SettingWithCopy warnings. Second, define autoregressive predictors explicitly so that only acceleration lags, never current x, y, or z values, are included. The final report should be regenerated from a clean kernel after these corrections.

Appendix A. Data Dictionary / Codebook

Variable	Role / type	Definition
Date	Timestamp	Observation time. The notebook converts it to an integer; use one consistent unit.
x [m/s2]	Numeric target	Acceleration error/measurement on the x-axis (m/s²).
y [m/s2]	Numeric target	Acceleration error/measurement on the y-axis (m/s²).
z [m/s2]	Numeric target	Acceleration error/measurement on the z-axis (m/s²).
T x [C]	Numeric predictor	Internal x-axis temperature (°C).
T y [C]	Numeric predictor	Internal y-axis temperature (°C).
T z [C]	Numeric predictor	Internal z-axis temperature (°C).
bottom x1 side [C]	Numeric predictor	Temperature at bottom x1 side (°C).
bottom x2 side [C]	Numeric predictor	Temperature at bottom x2 side (°C).
x sensor face [C]	Numeric predictor	Temperature at x sensor face (°C).
y sensor face [C]	Numeric predictor	Temperature at y sensor face (°C).
x elect face [C]	Numeric predictor	Temperature at x electronics face (°C).
y ACQ face [C]	Numeric predictor	Temperature at y acquisition face (°C).
top [C]	Numeric predictor	Temperature at top of package (°C).
inside air [C]	Numeric predictor	Internal air temperature (°C).
external [C]	Numeric predictor	External temperature (°C).
Altitude [m]	Numeric / excluded	Altitude in meters; removed before modeling in the notebook.
error_array	Derived target	Three-element target ordered [x, y, z].
*_lag_1/2/5/10	Engineered predictor	Prior values at 1, 2, 5, or 10 samples.
*_dT	Engineered predictor	One-sample first difference in a temperature channel.
temp_mean/std/range	Engineered predictor	Cross-sensor temperature summary at each time.
x_sensor_diff	Engineered predictor	x sensor-face temperature minus x electronics-face temperature.
y_sensor_diff	Engineered predictor	y sensor-face temperature minus y acquisition-face temperature.
hour/day/sin/cos	Engineered predictor	Calendar and cyclical time features used by the LSTM.

Appendix B. Notebook-to-Report Audit

- Source reviewed: basline_models.ipynb (36 cells; saved outputs used for reported values).
- Timestamp handling and uninterrupted-run selection are consistent in this notebook version: the longest run contains 746,265 observations.
- No Durbin-Watson, Ljung-Box, residual plots, or per-axis test metrics were present in the saved notebook.
- The autoregressive feature construction includes contemporaneous acceleration columns and must be corrected to lag-only inputs before its metrics are treated as valid.
- MSE values printed as 0.000000 for KNN/RF were rounded too aggressively; the report emphasizes MAE and R^2 instead of treating MSE as zero.

- Author name, submission date, and GitHub repository URL remain placeholders because they were not present in the supplied notebook or prompt.